

Muhammad Shoaib (PMP)[®]

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Technical Skills

- **AI Agent Development:** LLM-based automation, multi-agent systems, LangGraph, workflow orchestration, LinkedIn data extraction
- **Programming:** Python, R
- **ML & Data:** Machine Learning, Pandas, NumPy, PyTorch
- **Application Development:** PyQt / Tkinter (desktop), Django (web), WeChat Mini Program, MySQL
- **LLM API Integration:** Anthropic Claude, OpenAI GPT-4, Google Gemini, Alibaba Cloud Qwen

Education

Northwestern Polytechnical University (NPU) B.Sc. Computer Science & Technology Sep 2020 – Jul 2024
Xi'an, China

- Applied SVM and Random Forest classifiers to fMRI and MRI data to improve schizophrenia diagnostic accuracy and efficiency.
- Implemented a rigorous 10-fold cross-validation pipeline, achieving robust discrimination between schizophrenia patients and healthy controls.

Northwestern Polytechnical University (NPU) M.Sc. Computer Science & Technology Sep 2024 – Mar 2027
Xi'an, China (In Progress)

- Developed BAT 2.0, an open-source Python toolbox integrating Allen Human Brain Atlas transcriptomic data with CIBERSORTx deconvolution to perform single-cell-level cell-type annotation of neuroimaging results across 16 cell types.
- Designed statistical null models for multiple input types to correct for spatial and topological autocorrelation in cluster maps, parcellation atlases, and functional connectivity matrices.
- Validated the framework on 11 neurological and psychiatric disease datasets and one lesion-based benchmark derived from patient data.

Work Experience

Nanjing XinRui AI Technology Co., Ltd. AI Homework Supervision System Engineer Jun 2026 – Present

- Built a real-time student behavior monitoring system using computer vision and LLMs, integrating MediaPipe facial landmark detection, YOLOv8 object detection, and InsightFace ArcFace recognition to identify distraction, fatigue, poor posture, mobile phone use, and impersonation.
- Developed a Python FastAPI backend serving a real-time web dashboard and a WeChat Mini Program with bilingual (Chinese/English) interface support.
- Implemented an AI coaching module powered by DeepSeek LLM combined with the Pomodoro technique, delivering personalized TTS voice reminders and study motivation in real time.

Tech Stack: Python · FastAPI · OpenCV · MediaPipe · YOLOv8 · InsightFace ArcFace · DeepSeek LLM · Edge-TTS · WeChat Mini Program · SQLite

Shanghai Yugou Technology Co., Ltd. AI Engineer Intern Jun 2025 – Dec 2025

- Trained transfer-learning-based image classification models (EfficientNet) for pet emotion recognition, breed identification, and behavior classification; extended to audio-based emotion recognition from cat and dog vocalizations. Deployed all models into a pet health detection web application.
- Fine-tuned EfficientNet B0 for food image recognition, packaged the model for embedded chip deployment, and achieved real-time food identification from camera feeds with automated nutritional summaries (protein, carbohydrates, fat).
- Deployed YOLOv7 on an RK3588 edge platform for real-time small-animal detection and target tracking; integrated LiDAR with SLAM-based indoor mapping and autonomous obstacle avoidance; implemented PID motor control via STM32 for smooth target-following motion.
- Developed a WeChat Mini Program integrating pet emotion and health detection models, enabling users to

obtain pet status analysis from a single photograph.

Independent Project AI Agent – LinkedIn Profile Extractor Jan 2026 – Mar 2026

- Built an LLM-powered AI Agent that automatically extracts and structures LinkedIn profile data based on user-defined skills, departments, and geolocation criteria, integrating automated web scraping, natural language instruction parsing, and structured data output.

Independent Project Multi-Agent Research Report Generator Sep 2025 – Dec 2025

- Designed a five-agent collaborative system using LangGraph, automating the full pipeline from a user research query to a structured Markdown report. Agents handle orchestration, live web search (Tavily API), summarization, critique (up to two feedback loops), and writing.
- Integrated ChromaDB vector database for persistent memory; delivered via an interactive Streamlit web interface.

Tech Stack: Python · LangGraph · Alibaba Cloud DashScope API (Qwen) · Tavily · ChromaDB · Streamlit

Sage Education Institute Ltd. IT Technical Support Intern Jun 2022 – Aug 2022

- Provided system troubleshooting and diagnostic support to maintain operational efficiency; managed network operations and optimized system connectivity and performance.
- Implemented cybersecurity measures, updated security protocols, and led an internal software upgrade that improved functionality and user experience; reduced system vulnerabilities through enhanced security protocols.

Professional Profile

Computer Science professional with expertise in Python, machine learning, statistical analysis, desktop application development, and AI agent engineering. Track record of delivering production-ready systems combining LLMs, computer vision, and edge deployment. Strong mathematical foundation with hands-on experience building automated workflows and intelligent data extraction pipelines. Multilingual (English, Mandarin Chinese, Urdu/Punjabi); PMP-certified project manager.

Certifications & Languages

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| • NPU Outstanding Graduate (2024) | • Project Management Professional (PMP) |
| • NPU Future AI Masters International Summer Camp | • English: Duolingo English Test (130/160) |
| • University of Michigan: Programming for Everybody (Python) | • Chinese: HSK Level 3 (Intermediate) |
| • University of Michigan: Data Structures | • Skillvention Python Programming Certification |